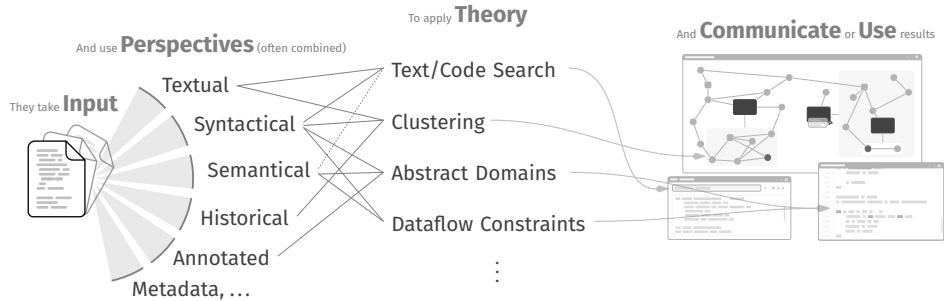




Static Analysis in the Real World

Software Quality Assurance — Static Code Analysis, III | Florian Sihler | December 17, 2025

Outline

1. A Small Recap

2. Introduction

3. Real-World Static Analyzers

4. Conclusion



Hans Houdini, mag Kraulen mehr als Abstract Interpretation



Sieglinde, vollzeit-cutie, hoch-motiviert

1. A Small Recap

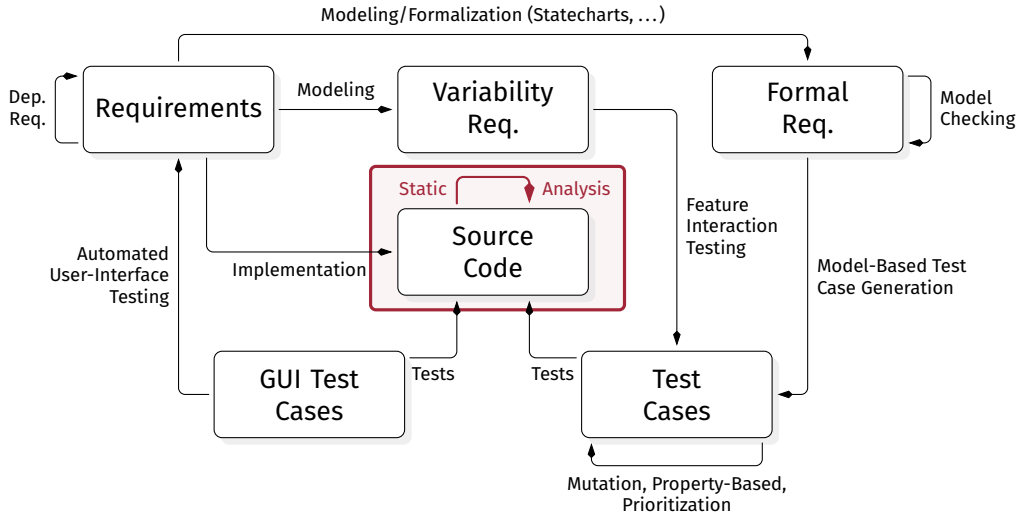
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Embedding a Landscape





What is static analysis?

Discover *syntactic/semantic properties* of programs
without running them.^[RY20]

What is abstract interpretation (in this context)?

A theory and framework to *systematically* derive sound
static analyses by **abstracting** program semantics.^[Cou21]



A way to soundly over-approximate all possible program behaviors.

On the Bread and Butter of Static Analysis

1. What are examples for semantic properties?
2. How do we describe semantic properties of programs?
3. What is a (complete) lattice, why do we need it?
4. What is described by a Galois connection?
5. Given a closed interval domain $[l..u]$, what are the abstract semantics of addition?

On the Bread and Butter of Static Analysis

1. What are examples for semantic properties?

“We never dereference null pointers”, “variable x is always positive”, “array a is sorted at a given point”, “we never divide by zero”, ...

2. How do we describe semantic properties of programs?

We use (po-)sets containing all program states that satisfy the property.

3. What is a (complete) lattice, why do we need it?

A poset, in which all subsets have a least upper bound (join), a greatest lower bound (meet), as well as a top and bottom element. We need lattices to describe how to combine properties (e.g., at control-flow joins) and to ensure that our analysis terminates.

4. What is described by a Galois connection?

Mapping concrete program states to abstract properties (abstraction, α) and back (concretization, γ).

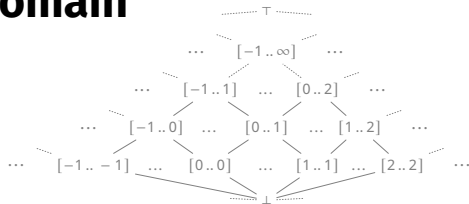
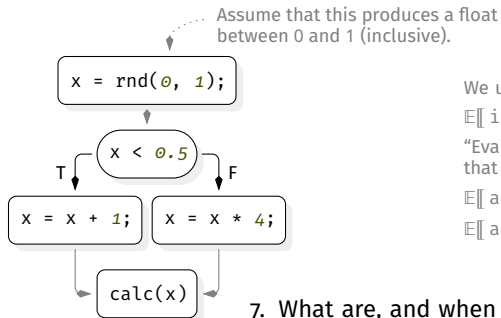
5. Given a closed interval domain $[l .. u]$, what are the abstract semantics of addition?

$$\mathbb{E} \llbracket a + b \rrbracket \rho \stackrel{\text{def}}{=} [l_a + l_b .. u_a + u_b] \quad (a \in [l_a .. u_a], b \in [l_b .. u_b])$$

*“We add the lower bounds and the upper bounds, assuming the environment ρ .”
(assuming a language like Java and ignoring overflows)*

Recap: Applying the Interval Domain

6. Do we need widening (∇) for this domain?



We use these semantics (use your intuition for the rest):

$$\llbracket \text{if}(c) a \text{ else } b \rrbracket \rho \stackrel{\text{def}}{=} \llbracket a \rrbracket (\llbracket c \rrbracket \rho) \sqcup \llbracket b \rrbracket (\llbracket \neg c \rrbracket \rho)$$

“Evaluate a assuming that c is true, evaluate b assuming that c is false, and join the results.”

$$\llbracket a + b \rrbracket \rho \stackrel{\text{def}}{=} [l_a + l_b .. u_a + u_b]$$

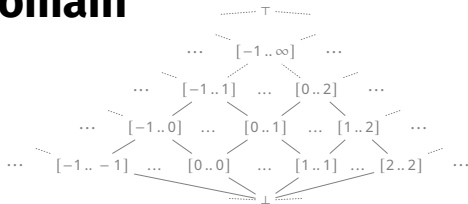
$$\llbracket a * b \rrbracket \rho \stackrel{\text{def}}{=} [\min S .. \max S] \text{ with } S = \{ l_a l_b, l_a u_b, u_a l_b, u_a u_b \} \\ (a \in [l_a .. u_a], b \in [l_b .. u_b])$$

7. What are, and when do we need fixpoints?

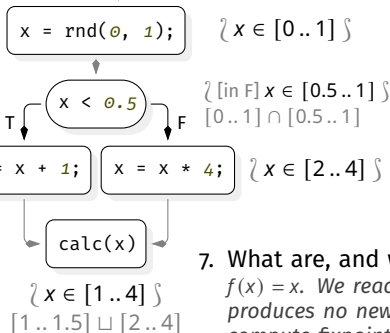
Recap: Applying the Interval Domain

6. Do we need widening (∇) for this domain?

Yes, because the lattice contains infinite ascending chains, e.g., $[0..0] \sqsubseteq [0..1] \sqsubseteq [0..2] \sqsubseteq \dots$



Over-Approx. to ignore the 0.499... problem
 $\{ [in\ T] x \in [0..0.5] \}$
 $[0..1] \cap [0..0.5]$



We use these semantics (use your intuition for the rest):

$$\llbracket \text{if}(c) a \text{ else } b \rrbracket \rho \stackrel{\text{def}}{=} \llbracket a \rrbracket (\llbracket c \rrbracket \rho) \sqcup \llbracket b \rrbracket (\llbracket \neg c \rrbracket \rho)$$

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7. What are, and when do we need fixpoints?

$f(x) = x$. We reach fixpoints when the repeated application of semantics produces no new info. When analyzing loops or recursive functions, we compute fixpoints to find a stable state (of all code executions).

2. Introduction

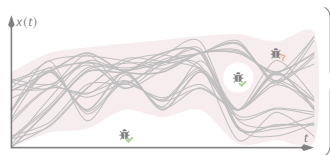
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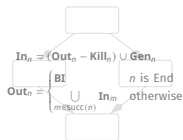
4. Conclusion

What we have... Theory



Abstractions

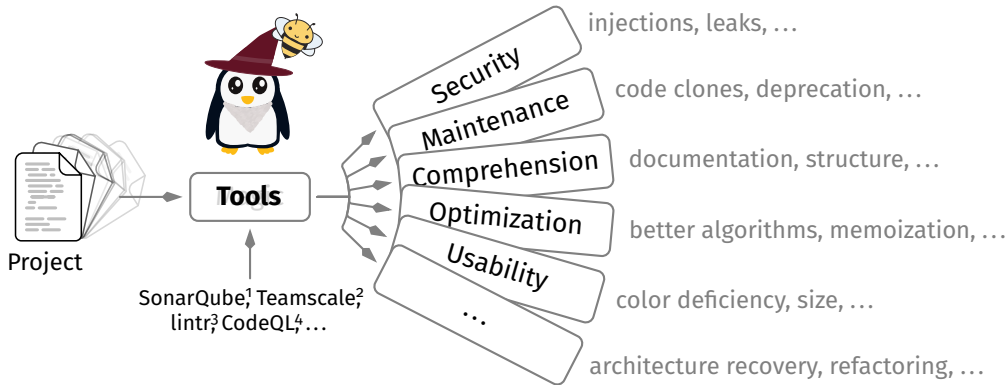
Abstractions



Data Flow Analysis

...

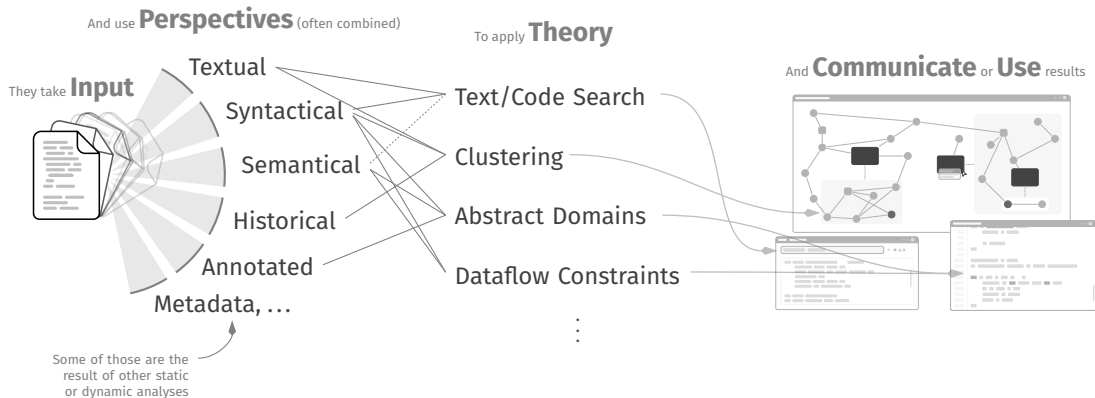
What we want... Tools



“Any sufficiently advanced technology is indistinguishable from magic.” — Arthur C. Clarke

¹ sonarsource.com, ² teamscale.com, ³ linter.r-lib.org, ⁴ codeql.github.com

What do they... do?



3. Real-World Static Analyzers

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4. Conclusion

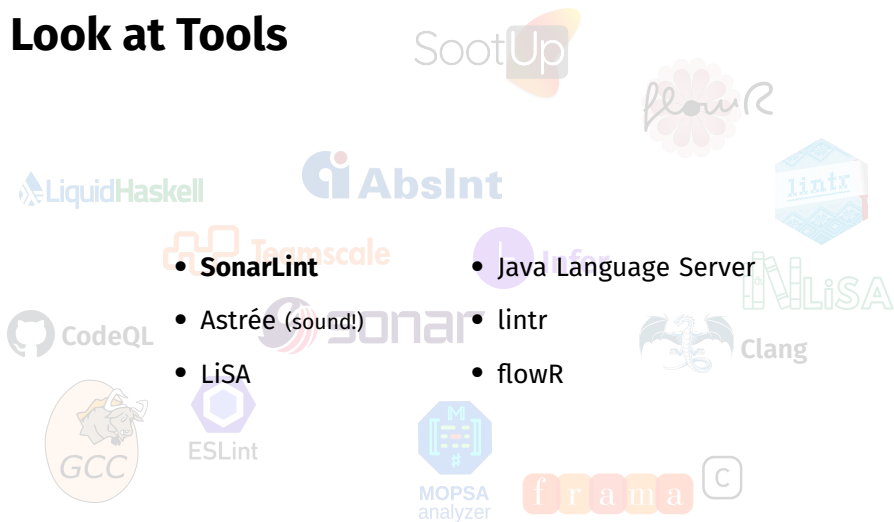
Let's Look at Tools



There are countless...

github.com/analysis-tools-dev/static-analysis

Let's Look at Tools



• **SonarLint**

• **Java Language Server**

• **Astrée (sound!)**

• **linter**

• **LiSA**

• **flowR**



ESLint



MOPSA
analyzer



There are countless...

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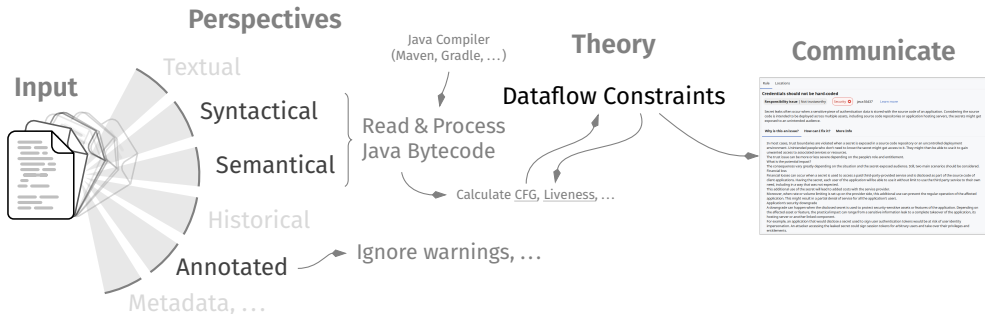
SonarLint



- Support for multiple languages (15+)
- Widely varying support and rules
- We focus on **sonar-java**

Separate frontends and analyzers per language!

733 rules!



SonarLint



- Support for multiple languages (15+)

- Widely varying support and rules

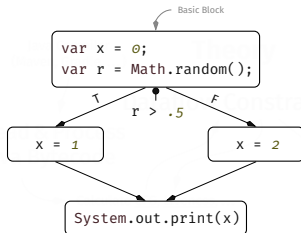
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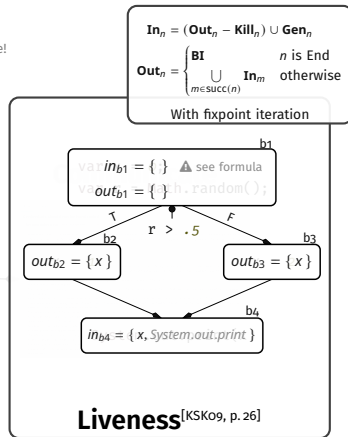
733 rules!

```
var x = 0;
var r = Math.random();
if(r > .5) {
    x = 1;
} else {
    x = 2;
}
System.out.print(x);
```

Code



Control Flow Graph



Liveness [KSK09, p. 26]

SonarLint — Unused Assignments



- Analyze “Dead Stores” (in 312 loc / 260 cloc):

```
int x = 0;  
x = 42;
```

- Use liveness analysis to obtain out_n of each basic block in the CFG
- Check if assignments ($x =$, $x++$, ...) are in out_n and resolved
- Check overwrites in the same basic block
- Minor special handling for **try-finally** blocks, ...

SonarLint — Unused Assignments



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- Use liveness analysis to obtain out_n of each basic block in the CFG
- Check if assignments ($x =$, $x++$, ...) are in out_n and resolved
- Check overwrites in the same basic block

- Minor special

```
77 LiveVariables liveVariables = LiveVariables.analyze(cfg);  
78 // Liveness analysis provides information only for block boundaries,  
   so we should do analysis between elements within blocks  
79 for(CFG.Block block : cfg.blocks()) {  
80     checkElements(block, liveVariables.getOut(block), methodSymbol);  
81 }
```

SonarLint — Hardcoded Credentials



- It does not always have to be that heavy! (116 loc / 87 cloc may suffice)
- For example, to identify hardcoded credentials:
`new PasswordAuthentication("password", "secret".toCharArray());`
- Traverse the Abstract Syntax Tree (AST)
- Check calls against a long list of signatures (currently 7664) with problematic indices
- Check if the arguments are “constant”
visiting the dataflow links and checking for predefined “plain text”

SonarLint — Hardcoded Credentials

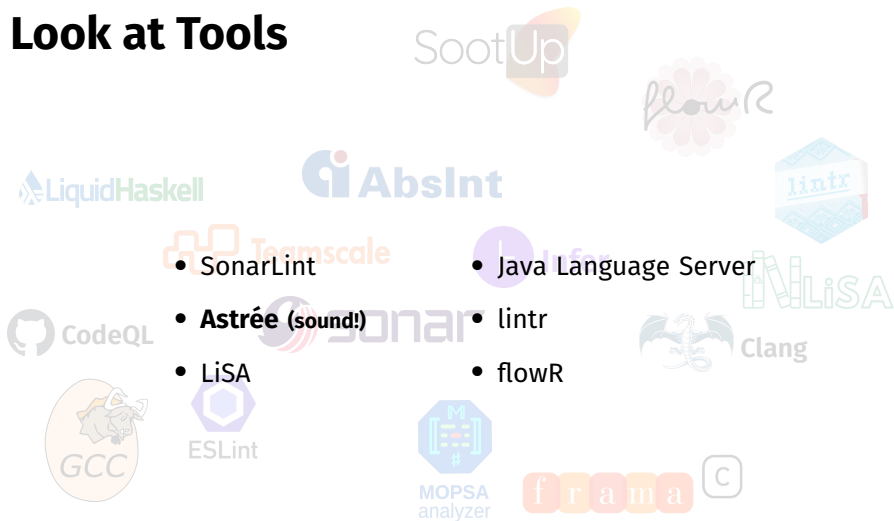


- It does not always have to be that heavy! (116 loc / 87 cloc may suffice)
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`new PasswordAuthentication("password", "secret".toCharArray());`
- Traverse the Abstract Syntax Tree (AST)
- Check calls against a long list of signatures (currently 7664) with problematic indices

• Check for hardcoded credentials

```
107 for (int targetArgumentIndex : method.indices) {  
108     ExpressionTree argument = arguments.get(targetArgumentIndex);  
109     var secondaryLocations = new ArrayList<JavaFileScannerContext.Location>();  
110     if (isExpressionDerivedFromPlainText(argument,  
111         secondaryLocations, new HashSet<>())) {  
112         reportIssue(argument, ISSUE_MESSAGE, secondaryLocations, null);  
113     }  
}
```

Let's Look at Tools



- SonarLint

- Java Language Server

- **Astrée** (sound!)

- lintr

- LiSA

- flowR



ESLint



MOPSA
analyzer

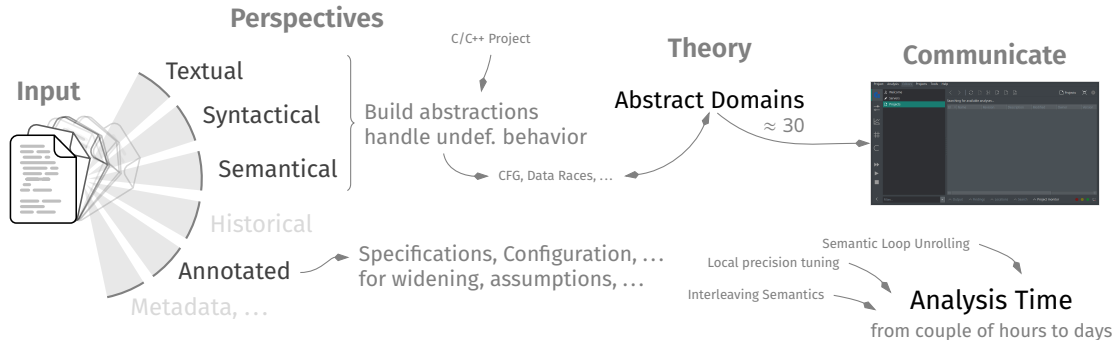


There are countless...

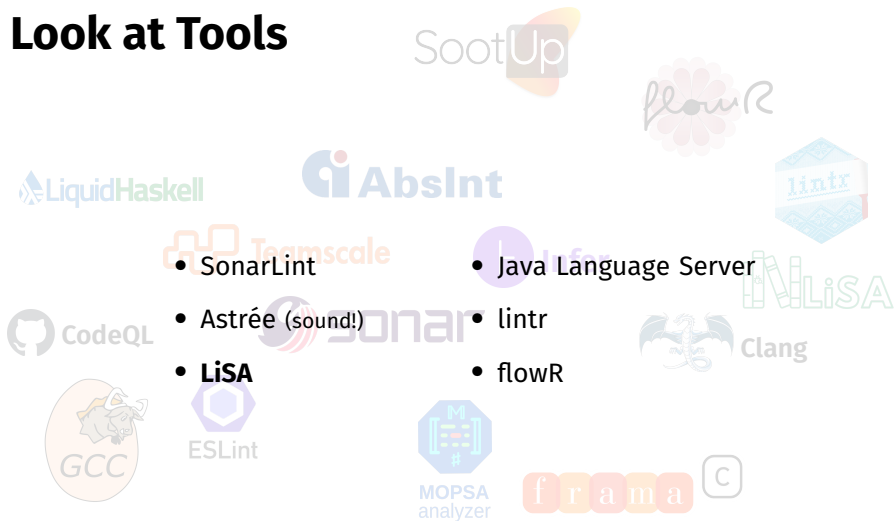
github.com/analysis-tools-dev/static-analysis

Astrée

- *Analyseur statique de logiciels temps-réel embarqués*
Static analyzer for real-time embedded software
- Proprietary, soundy static analyzer for C/C++
100+ directives and intrinsics, 140+ options
250 kloc of Ocaml, 240 kloc C/C++
- Uses abstract domains for timing validation, buffer overflows, ...



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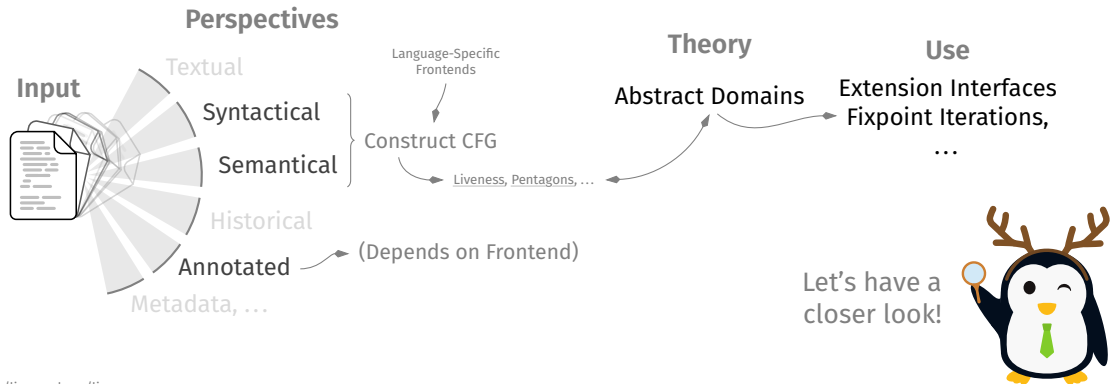
github.com/analysis-tools-dev/static-analysis

LiSA



- (Largely) Language Independent Library for Static Analysis
- Custom frontends for Rust, Go, Python, EVM, ...

Similar to Mopsa or Apron



LiSA — Interval Analysis^[Cou21, p. 389]

 `lisa-analyses/src/main/java/it/unive/lisa/analysis/numeric/Interval.java` (simplified)

```
57 Interval TOP = new Interval(IntInterval.INFINITY);
62 Interval BOTTOM = new Interval(null);

273 public Interval lubAux(Interval other) {
276     var newL = getLow().min(other.getLow());
277     var newH = getHigh().max(other.getHigh());
278     return new Interval(newLow, newHigh);
279 }

282 public Interval glbAux(Interval other) {
284     var newL = getLow().max(other.getLow());
285     var newH = getHigh().min(other.getHigh());
287     if(newLow.compareTo(newHigh) > 0) return bottom();
289     return new Interval(newLow, newHigh);
290 }
```

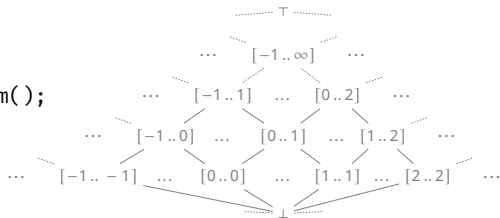
Widening, Narrowing, Assume, Satisfies, ...

$\top = [-\infty .. \infty]$ Top

$\perp = \emptyset$ Bottom

$\bigsqcup_k [\ell_k .. h_k] = [\min(\ell_k) .. \max(h_k)]$ Join

$\bigsqcap_k [\ell_k .. h_k] = [\max(\ell_k) .. \min(h_k)]$ Meet

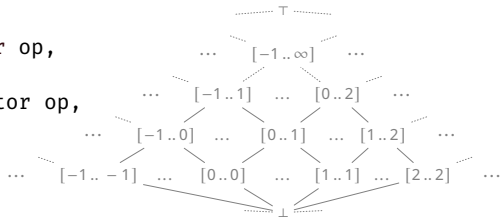


When to create which interval?

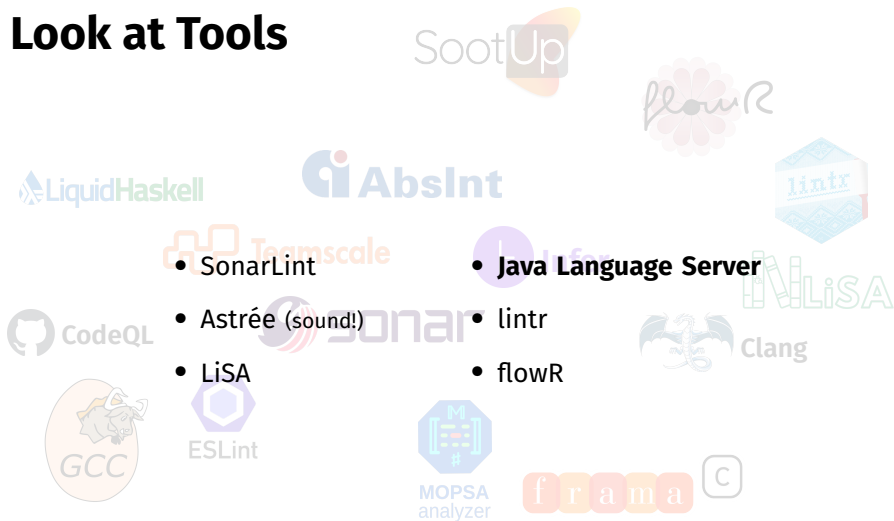
 [lisa-analyses/src/main/java/it/unive/lisa/analysis/numeric/Interval.java](#) (simplified)

```
144 public Interval evalNonNullConstant(Constant constant,  
146     ProgramPoint pp, SemanticOracle oracle) {  
148     if(constant.getValue() instanceof Integer) {  
149         var i = (Integer) constant.getValue();  
150         return new Interval(i, i);  
151     }  
152     return top();  
153 }  
  
157 public Interval evalUnaryExpression(UnaryOperator op,  
158     Interval arg, ...) { ... }  
188 public Interval evalBinaryExpression(BinaryOperator op,  
189     Interval left, Interval right, ...) { ... }
```

Fold-like Evaluation



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github.com/analysis-tools-dev/static-analysis

Java Language Server



- Uses the Language Server Protocol to provide static analysis for Java

Code Actions

```
public static void main(String[] args) {  
    foo();  
    foo  
}
```

Rename Refactoring

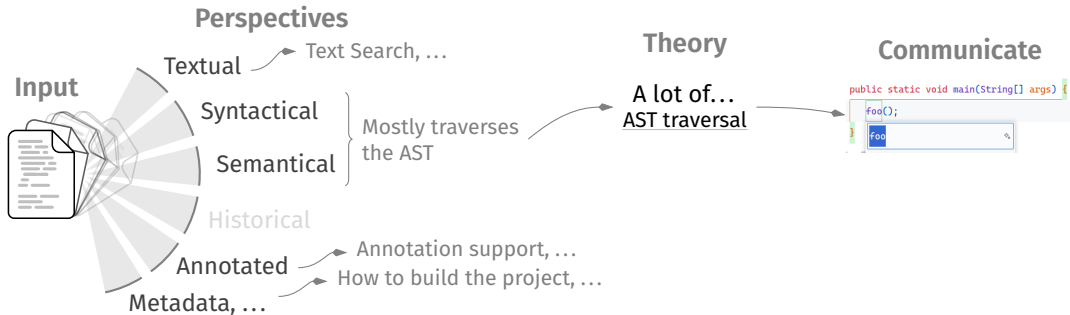
Go to Definition	F12
Go to Declaration	
Go to Type Definition	
Go to Implementations	Ctrl+F12
Go to References	Shift+F12
Go to Super Implementation	
Go to Test	
Peek	>
...	
Find All References	Shift+Alt+F12
Find All Implementations	
Show Call Hierarchy	Shift+Alt+H
Show Type Hierarchy	

- Relies on the Eclipse JDT Language Server

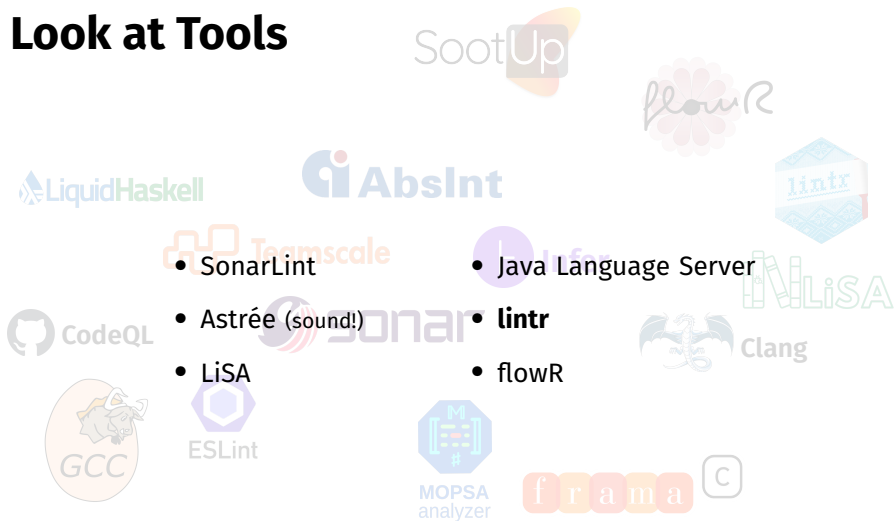


Java Language Server

- Uses the Language Server Protocol to provide static analysis for Java Renaming, Code Actions, ...
- Relies on the Eclipse JDT Language Server



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lintr



Why is this... special?

- A linter for the R programming language

```
x <- 4
f <- function() x
body(f) <- quote(y)
y <- 42
f() # 42
```

```
'if' <- function(...) 42
if(TRUE) print(3) # 42
```

```
x <- 2
'<-' <- '*'
x <- 21 # 42
```

- Common static analysis strategies have their... problems with R
- Most of R's users are no computer scientists (just a small set of existing work)
- So... how does *lintr* do it?
 - ~~Dataflow Constraints?~~
 - ~~Abstract Domains?~~
 - ~~Control Flow Graphs?~~
 - ~~AST Traversal?~~ (mostly) Pattern Matching and Evaluation!

lintr — Under the Hood

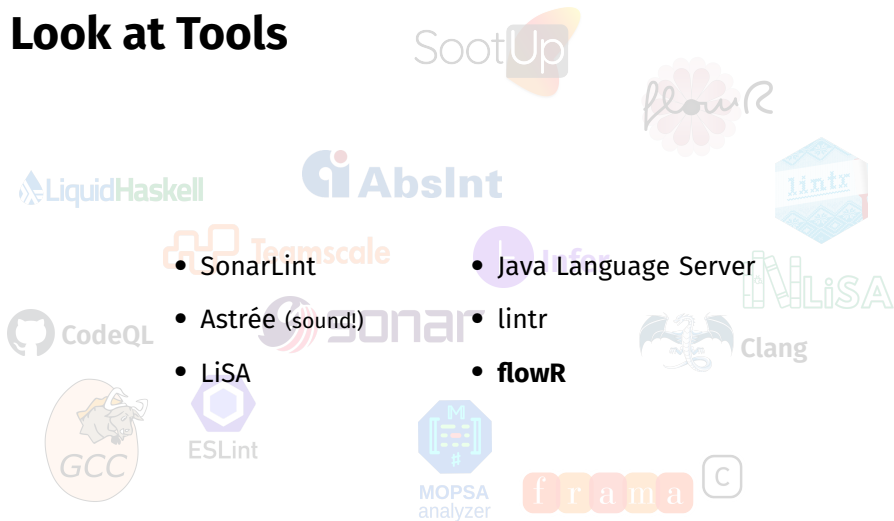
≈ 100 rules

 R/object_usage_linter.R

```
magic <- "  
  expr[LEFT_ASSIGN or EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]  
  | expr_or_assign_or_help[EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]  
  | equal_assign[EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]  
  | //SYMBOL_FUNCTION_CALL[text() = 'assign']/parent::expr/following-sibling::  
    expr[2][FUNCTION or OP-LAMBDA]  
  | //SYMBOL_FUNCTION_CALL[text() = 'setMethod']/parent::expr/following-sibling::  
    expr[3][FUNCTION or OP-LAMBDA]"
```

- What does this do?
This XPATH expression matches assignments to functions
- It is very rigid (no alias tracking, flow sensitivity, ...)
- And it... cheats:
try_silently(eval(envir = env, parse(text = code, keep.source = TRUE)))
It simply runs (parts of) the program (including side-effects), ...

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Existing Work

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- A static analysis framework for R

- Developed here, at Ulm University 

Florian Sihler, Julian Schubert, Oliver Gerstl, Lars Pfrenger, Johanna Scheck, Felix Schlegel, Ruben Dunkel, Thomas Schöller, Tim Schmidt, ...

- Let's get back to R:

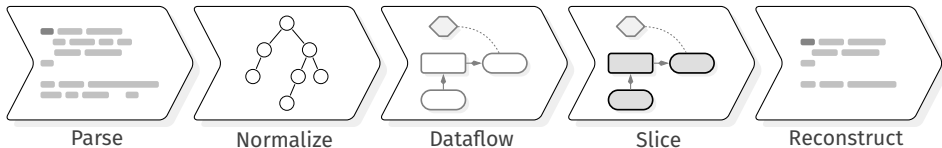
```
x <- 4  
f <- function() x  
body(f) <- quote(y)  
y <- 42  
f() # 42
```

```
'if' <- function(...) 42  
if(TRUE) print(3) # 42
```

```
x <- 2  
'<-' <- '*'  
x <- 21 # 42
```

- We have to intertwine dataflow- and control-flow analysis...

flowR — Architecture



```
sum  <- 0  
prod <- 1  
n    <- 10
```

```
for (i in 1:(n-1)) {  
  sum <- sum + i  
  prod <- prod * i  
}
```

```
cat("Sum:", sum, "\n")  
cat("Product:", prod, "\n")
```

slice(10, **sum**)

→

```
sum  <- 0  
prod <- 1  
n    <- 10
```

```
for (i in 1:(n-1)) {  
  sum <- sum + i  
  prod <- prod * i  
}
```

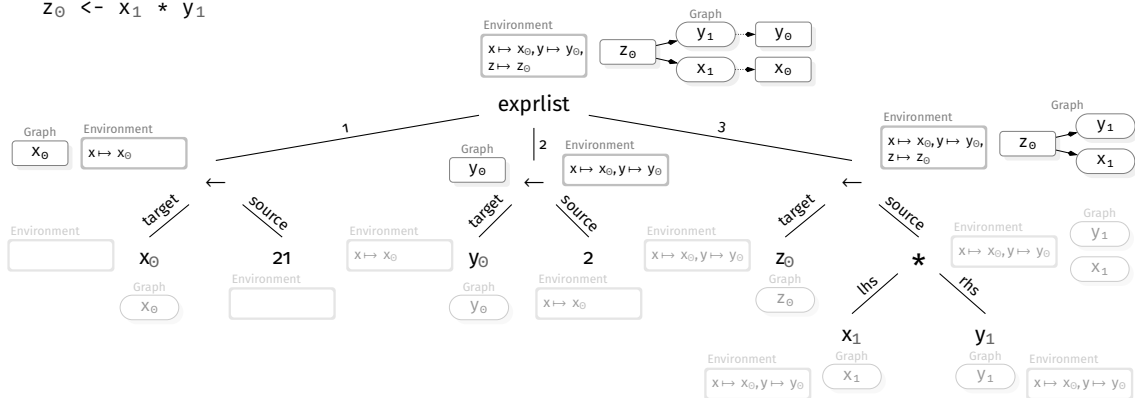
```
cat("Sum:", sum, "\n")  
cat("Product:", prod, "\n")
```

flowr — Dataflow

$x_0 \leftarrow 21$

$y_0 \leftarrow 2$

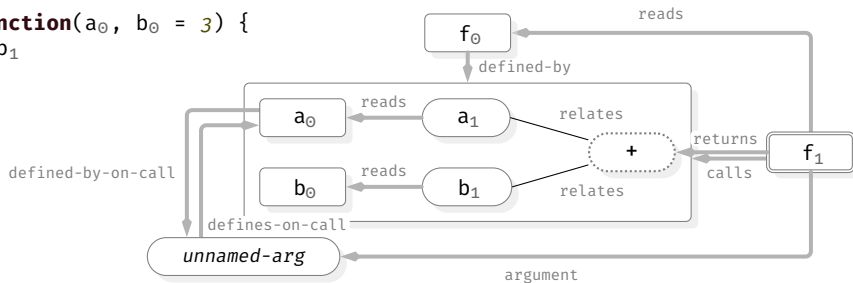
$z_0 \leftarrow x_1 * y_1$



Without Value tracking

flowR — There Is More...

```
f0 <- function(a0, b0 = 3) {  
  a1 + b1  
}  
f1(39)
```



flower



For brevity, we omit other requests

4. Conclusion

1. A Small Recap

2. Introduction

3. Real-World Static Analyzers

4. Conclusion

Soundness and Completeness Revisited

- We want to prove properties of programs (e.g., no overflow, shapes, ...)
- However, thanks to Rice [Ric53] we know:
Rice's theorem states that all nontrivial semantic properties of programs are undecidable. [Cou21, p. 100]

Soundness

- All properties we derive are true (but we may miss some)
- If we report bugs for violated properties, we produce no false negative

Completeness

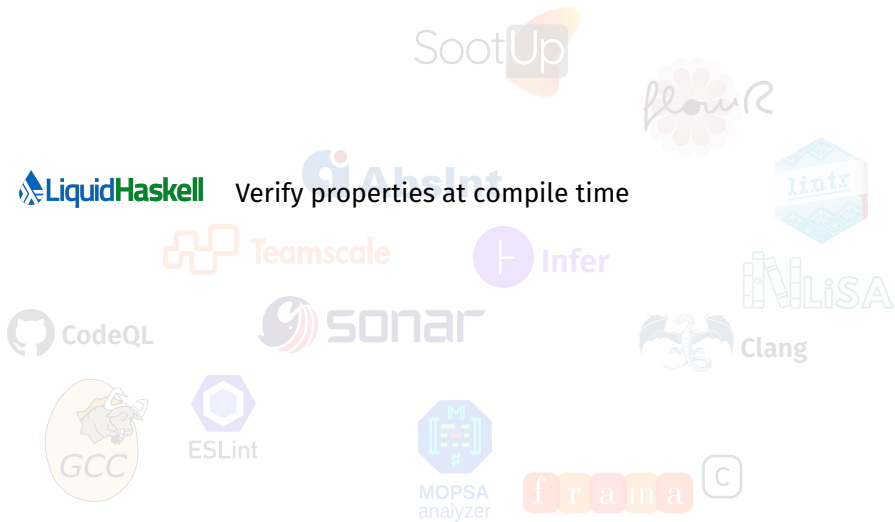
- We are able to infer all interesting properties in the program
- If we report bugs for violated properties, we produce no false positive



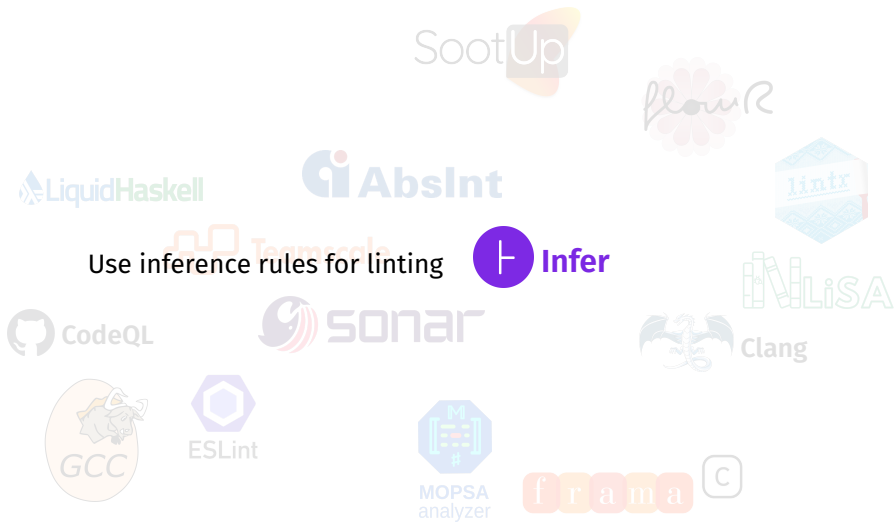
Handling Errors

- Analyzers tend to **favor soundness over completeness**
Better to report a false positive than to miss a bug
- Yet many cases are hard to handle (e.g., eval with *any* possible effect)
- Issues are usually ranked by severity (and sometimes confidence)
- Cutoffs may be applied if there are too many alarms
- There is still a lot of research required [EN08; Cou21, p. 705]

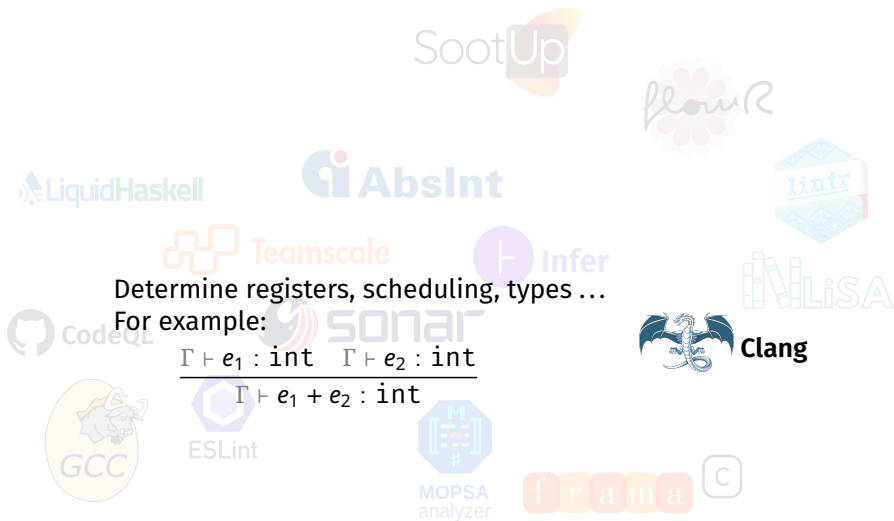
Oh There Are so Many Tools...



Oh There Are so Many Tools...



Oh There Are so Many Tools...



And There Are Many More Strategies

- **Static Analysis**
analyzing the code without executing it, for all possible runtime scenarios
- **Dynamic Analysis**
executing the program with specific input(s) to observe its behavior
- **Hybrid Analysis**
combining both strategies

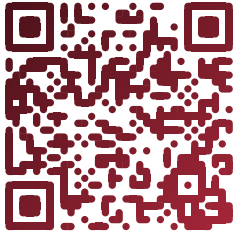
Program analysis is important. Additionally AND combined with testing.



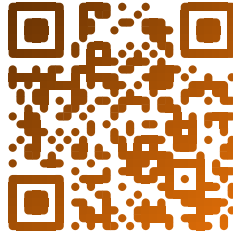
Back to the Questions

1. How would you capture what a *property* is? ✓
2. How would you phrase that one property is “better” than another? ✓
3. For what operations would you *not* use a control-flow graph? ✓
4. Why can't there be a fully automatic, sound, and complete static analyzer for general programs? ✓
5. What (big) additional challenges do you see in the real-world? ✓
6. How would you describe soundness and completeness? ✓
7. How would you describe Abstraction Interpretation? ✓

Thanksies!



slides and source code



feedback form

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